Supplement S2: The 33 Relations

Arithmetic, Dictionary Choices and Status Brieuc de La Fournière
Independent researcher, Beaune, France November 2026 | v3.6

S2.0 What this supplement establishes

No geometric map is known that sends the invariants of (K_7, $E_8 \times E_8$) to the Standard-Model parameters, and no rule assigns a formula to an observable (main §3.2). For each of the 33 relations (Type I in the dataset, S3), this supplement establishes three things:

1. **The arithmetic.** Given the declared constants, the closed form evaluates to the stated number. The arithmetic below is exact. Each relation has a Lean statement tying its closed form to its value (`relations_v36_certified`, Arithmon/K7-Lean); the form of each statement (exact rational equality, real closed form, bounds, integer) is given in S3.7.
2. **The dictionary choices.** Which constants the formula uses, which operations beyond integer ratios, and which items of the residue register it depends on (main §3.3, where the items R1 to R12 list what is set rather than derived).
3. **The status.** The status of each relation against the most recent data (main §4.1, generated from the dataset) is shown in the last column of each table, with its pull. The column is generated from the table of main §4.1 and is not edited here, so that the status has one source; the status vocabulary is defined in main §4.1.

The arithmetic of a relation says nothing about whether the formula is the right one for the observable (main §3.2, §7). Section S2.7 records the alternative formulas found in the project's record for the same observables; their number measures the vocabulary's freedom. Section S2.8 lists what changed since version 3.5.

S2.1 Notation

The twenty structural constants and their status (Lie datum, topological, input, combination) are listed in main §3.1. The relations also use the following, each a combination chosen by the framework:

Symbol	Value	Definition	Register
det(g)	65/32	$p_2 + 1/(b_2 + \dim G_2 - N_{gen}) = \det_{num}/\det_{den}$ (main App. B)	**imposed** normalization (register item R2); a second expression, Weyl · $\alpha_{sum}/2^{Weyl}$, agrees with it on the line $b_2 = 21$ and differs elsewhere
${\kappa_T}^{-1}$	61	$b_3 - \dim(G_2) - p_2$	integer combination without a physical name (register item R3); a second expression, $b_3 - b_2$ + Weyl, agrees with it on the line $b_2 = 21$ and differs elsewhere
φ	$(1 + \sqrt{5})/2$	through the McKay correspondence $E_8 \leftrightarrow 2I$	additional step (register item R9)
1	1	the integer 1	used freely as an additive constant
$2b_2$	42	also $p_2 \times N_{gen} \times \dim(K_7)$	combination
τ	3472/891	$\dim(E_8{\times}E_8) \cdot b_2/(\dim J_3(\mathbb{O}) \cdot H^*)$	used only in the dimensionful entries of the dataset (Type II, S3)

Weyl = 5. Three expressions give 5: $(\dim G_2 + 1)/N_{gen}$, $b_2/N_{gen} - p_2$, $\dim G_2 - \text{rank } E_8 - 1$. This is an arithmetic coincidence among the declared integers; it fixes no further quantity.

$N_{gen} = 3$. The integer identity $(\text{rank } E_8 + N_{gen}) \cdot b_2 = N_{gen} \cdot b_3$, i.e. $11 \cdot 21 = 3 \cdot 77$, is satisfied by $N_{gen} = 3$. It is a consistency check on (21, 77): no index theorem is known that produces it, and the expression $(77 - 8/3 \cdot 21)/7 = 3$ is the same arithmetic, with no operator whose index it would be (main §2.4).

S2.2 Gauge sector and Higgs

ID	Relation	Arithmetic	Choices declared	Status (main §4.1)
A01	$\sin^2\theta_W$ (M_Z, MS-bar)	$b_2/(b_3 + \dim G_2) = 21/91 = \mathbf{3/13} = 0.230769$	scale and scheme (R7)	integer; **proximity**; pull -7.5σ

ID	Relation	Arithmetic	Choices declared	Status (main §4.1)
A02	$\alpha_s(M_Z)$	$\sqrt{2}/(\dim G_2 - p_2) = \sqrt{\mathbf{2/12}} = 0.117851$	operation $\sqrt{}$; scale (R7)	algebraic; pull -0.2σ
A03	λ_H	$\sqrt{(\dim G_2 + N_{gen})/\det_{den}} = \sqrt{\mathbf{17/32}} = 0.128847$	operation $\sqrt{}$; det(g) (R2); compared with the tree-level $m_H{}^2/(2v^2)$	algebraic; det(g); pull -1.3σ
A04	$\alpha^{-1}(0)$	$(\dim E_8 + \text{rank } E_8)/2 + H^*/D_{bulk} + \det(g) \cdot \kappa_T = 128 + 9 + 65/1952 = \mathbf{267489/1952} = 137.03330$	det(g) (R2); κ_T (R3); scale $q^2 = 0$ (R7)	det(g), κ_T; **proximity**; pull $\gg 10\sigma$

On α^{-1}. The three terms carry no physical name: the first two are integer combinations, and the third is the product of the imposed normalization and the unnamed combination κ_T. The relation lies 19.7 ppm below $\alpha^{-1}(0)$, whose relative uncertainty is 1.5×10^{-10}: a proximity (main §4.2).

On λ_H. The comparison is with the tree-level $m_H{}^2/(2v^2)$, a scheme choice declared in the register (R7, main §3.3). A second formula for λ_H is on record (S2.7).

S2.3 Leptons (an overdetermined sector, register item R8)

ID	Relation	Arithmetic	Choices declared	Status (main §4.1)
A05	Q_{Koide}	$\dim G_2/b_2 = 14/21 = \mathbf{2/3}$	R8	integer; approximate (R8); pull $+0.4\sigma$
A06	m_τ/m_e	$\dim K_7 + 10 \cdot \dim E_8 + 10 \cdot H^* = 7 + 2480 + 990 = \mathbf{3477}$	R8; the coefficient 10 is not among the twenty constants	integer; approximate (R8); **tension**; pull -2.1σ

ID	Relation	Arithmetic	Choices declared	Status (main §4.1)
A07	m_μ/m_e	$\dim(J_3(\mathbb{O}))^\varphi = \mathbf{27}^\varphi = 207.0119$	R8; φ (R9)	φ; approximate (R8); **proximity**; pull $\gg 10\sigma$
A22	m_μ/m_τ	$(b_2 - D_{bulk})/\lvert PSL(2,7)\rvert = 10/168 = \mathbf{5/84} = 0.059524$	R8	integer; approximate (R8); **proximity**; pull $+20.8\sigma$

Four relations bear on two independent mass ratios, and they are mutually incompatible: with 27^φ and 3477, the Koide parameter is 0.666546, not 2/3. None is selected (main §6.1). The factorization $3477 = 3 \times 19 \times 61$ involves 19, which is not a structural constant. The associative 3-cycle volumes of S3 (the volume-difference entries of Type III) take $\ln(m_\tau/m_e) = 8.154$, the logarithm of the measured ratio, as their target, so their agreement with 3477 is the target returning and carries no independent information.

S2.4 Quarks, CKM and boson masses

ID	Relation	Arithmetic	Choices declared	Status (main §4.1)
A08	m_s/m_d	${p_2}^2 \times \text{Weyl} = 4 \times 5 = \mathbf{20}$	the PDG gives a range (17–22), not a measurement	integer; inside the PDG range
A09	m_c/m_s	$(\dim E_8 - p_2)/b_2 = 246/21 = \mathbf{82/7} = 11.7143$	n/a	integer; pull -1.6σ

ID	Relation	Arithmetic	Choices declared	Status (main §4.1)
A10	m_b/m_t	$1/(2b_2) = \mathbf{1/42} = 0.023810$	**mass scheme** (R7): m_b in MS-bar at m_b over the directly measured m_t	integer; **proximity** (in the scheme declared in R7); pull -10.2σ
A11	m_u/m_d	$(1 + \dim E_6)/\lvert PSL(2,7)\rvert = \mathbf{79/168} = 0.470238$	additive 1; the PDG quotes the uncertainty at 90% CL, divided by 1.645 to 1σ (S3)	integer; pull $+0.7\sigma$
A16	$\sin\theta_{12}$ (CKM) $= \lambda$	$\mathrm{fund}(E_7)/\dim E_8 = 56/248 = \mathbf{7/31} = 0.225806$	compared with $\sin\theta_{12}$, not $\sin^2\theta_{12}$; R12	integer; approximate (R12); pull $+0.9\sigma$
A17	A (Wolfenstein)	$(\mathrm{Weyl} + \dim E_6)/H^* = \mathbf{83/99} = 0.838384$	R12	integer; approximate (R12); pull $+0.7\sigma$
A18	$\sin\theta_{23}$ (CKM) $= A\lambda^2$	$\dim K_7/\lvert PSL(2,7)\rvert = 7/168 = \mathbf{1/24} = 0.041667$	compared with $\sin\theta_{23}$, not $\sin^2\theta_{23}$; R12	integer; approximate (R12); pull -0.3σ
A19	m_H/m_t	$\mathrm{fund}(E_7)/b_3 = 56/77 = \mathbf{8/11} = 0.727273$	m_t from direct measurements	integer; pull $+1.8\sigma$
A20	m_H/m_W	$(N_{gen} + \dim E_6)/\dim F_4 = \mathbf{81/52} = 1.557692$	n/a	integer; pull $+0.5\sigma$
A21	m_W/m_Z	$(2b_2 - \mathrm{Weyl})/(2b_2) = \mathbf{37/42} = 0.880952$	n/a	integer; **proximity**; pull -3.8σ

The CKM sector is overdetermined (register item R12). The three relations bear on two Wolfenstein parameters, since $\sin\theta_{23} = A\lambda^2$. With $\lambda = 7/31$ and $A = 83/99$, $A\lambda^2 = 4067/95139 = 0.042748$, while the $\sin\theta_{23}$ relation gives $1/24 = 0.041667$: the three relations are 2.6% apart as exact equalities, and at most two can hold. None is selected (main §4.2).

Sines, not squared sines. The two CKM relations 7/31 and 1/24 are compared with $\lambda = \sin\theta_{12}$ and $A\lambda^2 = \sin\theta_{23}$ (S3.3).

On the constant 42. The integer $42 = 2b_2$ appears in both $m_b/m_t = 1/42$ and $\Omega_{DM}/\Omega_b = (1 + 42)/8$. Two formulas that share an integer share a choice of vocabulary; no connection between the quantities is implied.

m_W/m_Z from the published masses. Like the other mass ratios, m_W/m_Z is computed from the PDG masses, 80.3625(77)/91.1879(20) = 0.88128(9), correlations neglected (S3.3).

Dependence noted, not counted. $\sin^2\theta_W$ in MS-bar and m_W/m_Z are related in the Standard Model through radiative corrections; the two relations are not independent tests in the strict sense.

S2.5 Neutrino mixing

ID	Relation	Arithmetic	Choices declared	Status (main §4.1)
A12	$\sin^2\theta_{12}$ (PMNS)	$(1 + N_{gen})/\alpha_{sum} = \mathbf{4/13} = 0.307692$	additive 1	integer; pull -0.2σ
A13	$\sin^2\theta_{23}$ (PMNS)	$(D_{bulk} - \text{Weyl})/D_{bulk} = \mathbf{6/11} = 0.545455$	octant (main §4.3)	integer; **tension** (octant; inside the NuFIT 3σ range, main §4.3); pull $+4.4\sigma$, not Gaussian (main §4.3)
A14	$\sin^2\theta_{13}$ (PMNS)	$D_{bulk}/\dim(E_8\times E_8) = \mathbf{11/496} = 0.022177$	n/a	integer; pull -0.5σ

ID	Relation	Arithmetic	Choices declared	Status (main §4.1)
A15	δ_{CP}	dim K_7 · dim G_2 + H* = 98 + 99 = **197**, read as the phase $2\pi \cdot 197/360$	**unit** (R6)	**unit assignment** (R6); pull -0.4σ

The δ_{CP} rule. The prediction is the phase $2\pi \cdot 197/360$, a point value; the unit is a dictionary assignment (register item R6), and the decision rule of main §11.1 for δ_{CP} applies: a measurement whose 3σ interval excludes 197° rejects the relation and the unit assignment together, with no repair clause and no theoretical tolerance added afterwards.

S2.6 Cosmology (an overdetermined sector, register item R11)

ID	Relation	Arithmetic	Choices declared	Status (main §4.1)
A23	$\Omega_{\mathrm{DE}} = \Omega_\Lambda$	ln 2 · (b_2 + b_3)/H* = ln 2 · 98/99 = **0.686146**	operation ln 2; R11	ln 2; approximate (R11); pull $+0.2\sigma$
A24	n_s	$\zeta(D_{\mathrm{bulk}})/\zeta(\mathrm{Weyl})$ = ζ**(11)**/ζ**(5)** = 0.964864	operation ζ	ζ values; pull 0.0σ
A25	$\Omega_{\mathrm{DM}}/\Omega_{\mathrm{b}}$	(1 + 2b_2)/rank E_8 = **43/8** = 5.375	additive 1; R11	integer; approximate (R11); pull $+0.1\sigma$
A26	$\Omega_{\mathrm{c}}/\Omega_\Lambda$	$\det_{\mathrm{num}}$/|PSL(2,7)| = **65/168** = 0.386905	R2; R11	integer; det(g); approximate (R11); pull 0.0σ

ID	Relation	Arithmetic	Choices declared	Status (main §4.1)
A27	Ω_Λ/Ω_m	$(\dim G_2 + H^*)/\dim F_4 = \mathbf{113/52} = 2.173077$	R11	integer; approximate (R11); pull 0.0σ
A28	h	$(\lvert PSL(2,7)\rvert - 1)/\dim E_8 = \mathbf{167/248} = 0.673387$	additive 1	integer; pull 0.0σ
A29	Ω_b/Ω_m	$\mathrm{Weyl}/\det_{den} = \mathbf{5/32} = 0.15625$	R2; R11	integer; det(g); approximate (R11); pull -0.1σ
A30	Ω_c/Ω_m	$(\det_{den} - \mathrm{Weyl})/\det_{den} = \mathbf{27/32} = 0.84375$	R2; R11	integer; det(g); approximate (R11); **tension**; pull $+2.5\sigma$
A31	σ_8	$(p_2 + \det_{den})/(2b_2) = 34/42 = \mathbf{17/21} = 0.809524$	R2	integer; det(g); pull -0.3σ
A32	Y_p	$(1 + \dim G_2)/{\kappa_T}^{-1} = \mathbf{15/61} = 0.245902$	additive 1; R3	integer; κ_T; pull $+0.3\sigma$
A33	Ω_m/Ω_Λ	$(1 + \dim J_3(\mathbb{O}))/{\kappa_T}^{-1} = \mathbf{28/61} = 0.459016$	additive 1; R3; R11	integer; κ_T; approximate (R11); pull -0.1σ

The seven density relations bear on two independent densities (flat ΛCDM: $\Omega_b + \Omega_c + \Omega_\Lambda = 1$) and are mutually incompatible as exact equalities (main §4.2):

Quantity	Values implied by the relations
Ω_Λ/Ω_m	113/52 = 2.1731 (the Ω_Λ/Ω_m relation); 61/28 = 2.1786 (inverse of the Ω_m/Ω_Λ relation); 2.1862 (Ω_{DE} with flatness)
Ω_c/Ω_b	43/8 = 5.375 (the Ω_{DM}/Ω_b relation); 27/5 = 5.400 (Ω_c/Ω_m over Ω_b/Ω_m)
Ω_c/Ω_Λ	65/168 = 0.3869 (the Ω_c/Ω_Λ relation); 0.3873 (Ω_c/Ω_m times Ω_m/Ω_Λ); 0.3883 (Ω_c/Ω_m over Ω_Λ/Ω_m)

Ω_Λ/Ω_m and Ω_m/Ω_Λ are the same quantity, written with two formulas. The disagreements are below the Planck uncertainties of these ratios, which is why six of the seven pass at 2σ; Ω_c/Ω_m, whose uncertainty is much smaller (0.0019), is in tension at $+2.5\sigma$. The seven cannot hold together; none is selected (R11). Some are redundant and some subsets are compatible: the Ω_b/Ω_m and Ω_c/Ω_m relations are equivalent when neutrinos are neglected ($\Omega_m = \Omega_b + \Omega_c$, so that 5/32 + 27/32 = 1), with or without flatness, and the largest compatible subsets have three relations, these two with one of Ω_{DE}, Ω_c/Ω_Λ, Ω_Λ/Ω_m or Ω_m/Ω_Λ (checked by solving each subset for (Ω_b, Ω_c)).

Not among the 33. $\Omega_m = \Omega_{DE}/\sqrt{\text{Weyl}} = 0.3068$ (2.7% from Planck) would be an eighth density relation, incompatible with the others; it is not part of the dataset. The factor ln 2 in Ω_{DE} is an operation of the grammar, with no computation behind it. Against the Planck 2018 chains, Ω_{DM}/Ω_b is 5.364 ± 0.077.

S2.7 Vocabulary freedom: alternative formulas on record

For each relation one can count the distinct expressions in the constants that reduce to the same value (for example 14 for $\sin^2\theta_W = 3/13$, 20 for $Q_{Koide} = 2/3$, 21 for $m_b/m_t = 1/42$). A value reached by many paths through twenty constants is a value the vocabulary reaches easily: the count measures the vocabulary's freedom (register items R4, the formula assignments, and R5, the grammar and vocabulary). It is not what the look-elsewhere test of main §7 prices: the published values 3/13 and 1/42 lie outside their 2σ windows, and the 30 targets that every competitor reaches are already reached by the universal constants of the grammar, with no constant of K_7 at all (S4.2.2). The counts are:

Relation	Distinct expressions recorded	Examples
$\sin^2\theta_W = 3/13$	14	N_{gen}/α_{sum}; $b_2/(b_3 + \dim G_2)$; $\dim J_3(\mathbb{O})/(\dim F_4 + \det_{num})$; $(1 + \dim G_2)/\det_{num}$
$Q_{Koide} = 2/3$	20	p_2/N_{gen}; $\dim G_2/b_2$; $\dim F_4/\dim E_6$; $\text{rank } E_8/(\text{Weyl} + \dim K_7)$

Relation	Distinct expressions recorded	Examples
$m_b/m_t = 1/42$	21	$1/(2b_2)$; $(1 + N_{gen})/\lvert PSL(2,7)\rvert$; $p_2/(\dim K_7 + b_3)$; $N_{gen}/(\dim J_3(\mathbb{O}) + H^*)$
$\sin^2\theta_{12}$ (PMNS) = 4/13	28	n/a
$m_H/m_t = 8/11$	19	n/a
$m_u/m_d = 79/168$, $m_H/m_W = 81/52$	1 each	n/a

Second formulas for the same observable. The project's record also contains formulas that give a *different* value for the same observable, chosen at different times:

Observable	Dataset formula	Other formula on record
λ_H	$\sqrt{17/32} = 0.12885$	$b_2(M_1)/(b_3 + b_2(M_2)) = 11/87 = 0.12644$ (TCS building blocks)
Ω_{DE}	$\ln 2 \cdot 98/99 = 0.68615$	$1 - \chi(K3)/b_3 = 53/77 = 0.68831$
m_c/m_s	$246/21 = 11.714$	$b_2(M_1) + \dim K_7/b_2(M_2) = 117/10 = 11.700$
Ω_Λ/Ω_m	$113/52 = 2.1731$	$(\det_{den} - \dim K_7)/D_{bulk} = 25/11 = 2.2727$; $61/28$ (the Ω_m/Ω_Λ relation inverted)
θ_{13}	$\sin^2\theta_{13} = 11/496$	$\theta_{13} = \pi/b_2 = 8.571°$; $\theta_{13} = 94/11 = 8.545°$ read in degrees
θ_{23}	$\sin^2\theta_{23} = 6/11$ ($\theta = 47.6°$)	$\theta_{23} = \arcsin(75/99) = 49.25°$ ($\sin^2\theta_{23} = 0.574$)
θ_{12}	$\sin^2\theta_{12} = 4/13$	$\theta_{12} = \arctan\sqrt{((2\pi/25)/(511/884))} = 33.40°$
δ_{CP}	197	$197 \times 62/69 = 177.01$ (S2.8)

The formulas labelled "TCS building blocks" (twisted connected sum) use $b_2(M_1) = 11$ and $b_2(M_2) = 10$, Betti numbers of building blocks whose identification is open (main §9). Several alternatives were introduced after a release of the data had moved: the Ω_{DE} alternative was preferred against one Planck release and not the other, and the θ_{13} alternative "in degrees" is a second unit assignment of the kind register item R6 records. None of these is part of the 33; they are listed because they show, on the framework's own record, how many formulas the vocabulary offers for one observable and how the choice among them was made.

Patterns. The Fibonacci and Lucas numbers among the constants (2, 3, 5, 8, 13, 21; 7, 11) and the identities among the integers ($\dim G_2 = p_2 \cdot \dim K_7$, $b_2 = N_{gen} \cdot \dim K_7$, $\lvert PSL(2,7)\rvert = \text{rank } E_8 \cdot b_2$, $b_3 + \dim G_2 = \dim K_7 \cdot \alpha_{sum}$) are arithmetic facts about the chosen integers. Small integers satisfy many such identities; no physical consequence follows from these.

S2.8 Changes since version 3.5

The sections above state the present position. This section lists, once, what versions up to 3.5 of this supplement said that version 3.6 no longer says.

Earlier statement	Present position
the supplement titled "Complete Derivations", each relation presented as a theorem with a proof	arithmetic, dictionary choices and status (S2.0)
"proof", "VERIFIED", "TOPOLOGICAL" as statuses of the relations	the arithmetic is exact; the assignment is not derived (S2.0)
$N_{gen} = 3$ by the Atiyah–Singer index theorem	no operator is specified; the expression is the integer identity of S2.1
"All 33 predictions use only topological invariants; no fitting"	eight use the imposed normalization or κ_T (R2, R3); the formulas were chosen with the values known (R4)
Mean deviation 0.73%, "11 exact matches", deviation histograms	computed against an earlier dataset; a mean deviation measures closeness, which is not distinctive (main §4.2, §7)
Statistical uniqueness: 192 349 alternatives, none better, $P < 5 \times 10^{-6}$, gauge-group and holonomy rankings	the alternatives were scored with the framework's own formulas, not allowed to search; replaced by the preregistered test of main §7 and S4
"Literature uniqueness" of (21, 77)	absence from a catalogue is not selection; the selection audit is main §10
Results "from analysis of the Newton–Kantorovich-certified metric": V_{min} formula, $U(1)^2$ symmetry, origin of the SM gauge group from triality	properties of an approximate local metric (main §2.3, §9); not used by any relation; the gauge group is assumed (main §2.3)
$\kappa_T = 1/61$ as a "topological torsion capacity"; $61 = \dim F_4 + N_{gen}{}^2$	an integer combination without a physical name (R3); a second expression for 61 is another instance of S2.7
$\det(g) = 65/32$ "consistent with the topological formula" via three algebraic paths	imposed normalization (R2); the metric optimization was constrained to this target
3477 "factorizes into framework constants"; a "period-integral cross-confirmation" of 3477 from associative volumes	19 is not a structural constant; the volumes take the measured ratio as their target (S2.3)
$\Omega_m = \Omega_{DE}/\sqrt{5}$ as a "derived" relation; $\Omega_{DM}/\Omega_b = 43/8$ "exact" (from a dataset value 5.375 ± 0.1 that matched no Planck column); ln 2 as a "binary information factor" of the moduli space	not among the 33; 5.364 ± 0.077 on the Planck chains; an operation of the grammar (S2.6)
the three terms of α^{-1} named "algebraic", "bulk impedance" and "torsional correction"	no physical name (S2.2)

Earlier statement	Present position
Weyl = 5 as a "triple identity"; the additive constant 1 written b_0	an arithmetic coincidence; the integer 1 (S2.1)
the two CKM relations labelled $\sin^2\theta_{12}$ and $\sin^2\theta_{23}$	sines (S2.4, S3.3)
42 in m_b/m_t and Ω_{DM}/Ω_b as "connecting quark physics to large-scale structure"	a shared choice of vocabulary (S2.4)
counts of equivalent expressions as "structural robustness"; an "algebraic web" and a "Fibonacci–Lucas embedding" reflecting icosahedral geometry	measures of the vocabulary's freedom; arithmetic facts (S2.7)
"at most two of the seven density relations can be exact" (earlier drafts of 3.6)	the largest compatible subsets have three (S2.6)

The factor 62/69 (Appendix F of earlier versions). A search after the NuFIT 6.0 release found $197 \times 62/69 = 177.01$, matching that release's central value, and earlier versions kept it as a "contingency plan": if DUNE measured δ_{CP} near 177°, the framework would adopt 177.01 and read 197 as a "bare" value. That is a repair clause, and the decision rule has none (main §11.1): if a measurement excludes 197° at 3σ, the relation $\delta_{CP} = 197$ and its unit assignment are rejected, and no factor found afterwards replaces them. The factor $62/69 = 248/276$ is recorded in S2.7 as one more formula the vocabulary offers.

References

[1] Particle Data Group, F. Takahashi et al., Int. J. Mod. Phys. A 41, 2630011 (2026)

[2] S. Weinberg, Phys. Rev. D 13, 974 (1976)

[3] Planck Collaboration, A&A 641, A6 (2020); chains: Planck Legacy Archive, COM_CosmoParams_base-plikHM-TTTEEE-lowl-lowE_R3.00

[4] A.G. Riess et al., ApJL 934, L7 (2022)

[5] C.D. Froggatt, H.B. Nielsen, Nucl. Phys. B 147, 277 (1979)

[6] Y. Koide, Lett. Nuovo Cim. 34, 201 (1982)

[7] C. Furey, *Standard Model Physics from an Algebra?*, PhD thesis, U. Waterloo (2015)

[8] C. Furey, M.J. Hughes, Phys. Lett. B 831, 137186 (2022)

[9] R. Wilson, arXiv:2404.18938 (2024)

[10] T.P. Singh et al., arXiv:2206.06911v3 (2024)

[11] B.S. Acharya, S. Gukov, Phys. Rep. 392, 121 (2004)

[12] L. Foscolo et al., Duke Math. J. 170, 3 (2021)

[13] A. Kovalev, "Coassociative K3 fibrations of compact G_2-manifolds," arXiv:math/0511150 (2005)

[14] M. Haskins, J. Nordstrom, arXiv:1809.09083 (2022)

[15] G. Bera, "Associative submanifolds in twisted connected sum G_2-manifolds," arXiv:2209.00156 (2022)

[16] D. Crowley, S. Goette, J. Nordstrom, Invent. Math. (2025)

[17] T. Dray, C.A. Manogue, *The Geometry of the Octonions*, World Scientific (2015)

[18] J.F. Adams, *Lectures on Exceptional Lie Groups*, U. Chicago Press (1996)

[19] D.J. Gross et al., Nucl. Phys. B 256, 253 (1985)

[20] D.D. Joyce, *Compact Manifolds with Special Holonomy*, Oxford U. Press (2000)

[21] A. Kovalev, J. Reine Angew. Math. 565, 125 (2003)

[22] A. Corti et al., Duke Math. J. 164, 1971 (2015)

[23] R. Harvey, H.B. Lawson, Acta Math. 148, 47 (1982)

[24] B.S. Acharya, Class. Quant. Grav. 19, 5619 (2002)

[25] B.S. Acharya, E. Witten, arXiv:hep-th/0109152 (2001)

[26] The NOvA Collaboration, The T2K Collaboration, "Joint neutrino oscillation analysis from the T2K and NOvA experiments," Nature 646, 818-824 (2025), doi:10.1038/s41586-025-09599-3, arXiv:2510.19888

[27] I. Esteban, M.C. Gonzalez-Garcia, M. Maltoni, I. Martinez-Soler, J.P. Pinheiro, T. Schwetz, "NuFit-6.0: updated global analysis of three-flavor neutrino oscillations," JHEP 12 (2024) 216, arXiv:2410.05380; NuFIT 6.1: three-neutrino fit based on data available in November 2025, www.nu-fit.org

[28] L. de Moura, S. Ullrich, CADE 28, 625 (2021)

[29] mathlib Community, github.com/leanprover-community/mathlib4

[30] DUNE Collaboration, FERMILAB-TM-2696 (2020)

[31] DUNE Collaboration, arXiv:2103.04797 (2021)

[32] E. Witten, "Strong coupling expansion of Calabi-Yau compactification," Nucl. Phys. B 471, 135 (1996)

[33] B.S. Acharya et al., Phys. Rev. D 76, 126010 (2007)

[34] M. Atiyah, E. Witten, Adv. Theor. Math. Phys. 6, 1 (2002)

[35] G. Kane, *String Theory and the Real World* (2017)

[36] J. Distler, S. Garibaldi, Commun. Math. Phys. 298, 419 (2010)

[37] J.C. Baez, "Octonions and the Standard Model," math.ucr.edu/home/baez/standard/ (2020-2025)

[38] Springer Nature, "Artificial intelligence (AI) policy," www.springernature.com/gp/policies (2024)

[39] J.A. Wheeler, "Information, physics, quantum: the search for links," in *Complexity, Entropy, and the Physics of Information* (W.H. Zurek, ed.), Addison-Wesley (1990), pp. 3–28.

[40] J. Worrall, "Structural realism: the best of both worlds?," *Dialectica* **43**, 99–124 (1989).

[41] J. Ladyman, D. Ross, *Every Thing Must Go: Metaphysics Naturalized*, Oxford University Press (2007).

[42] R. Pinčák, A. Pigazzini, M. Pudlák, E. Bartoš, "Geometric origin of a stable black hole remnant from torsion in G_2-manifold geometry," Gen. Rel. Grav. **58**, 29 (2026), doi:10.1007/s10714-026-03528-z

[43] D. Joyce, S. Karigiannis, "A new construction of compact torsion-free G_2-manifolds by gluing families of Eguchi-Hanson spaces," *J. Differential Geom.* (2021), arXiv:1707.09325 (2017).

[44] S. Mukai, "Finite groups of automorphisms of K3 surfaces and the Mathieu group," *Invent. Math.* **94**, 183–221 (1988).

[45] A. Garbagnati, A. Sarti, "Symplectic automorphisms of prime order on K3 surfaces," *J. Algebra* **318**, 323–350 (2007), arXiv:math/0603742.

[46] LEP2 SUSY Working Group (ALEPH, DELPHI, L3, OPAL), "Combined LEP chargino results, up to 208 GeV for large m0," note LEPSUSYWG/01-03.1 (2001), lepsusy.web.cern.ch.

[47] Fermilab News, "Fermilab marks major milestone for world-leading DUNE experiment," 7 May 2026, news.fnal.gov.

[48] J. Chen, H. Hong, "Intermediate curvature and splitting theorem," arXiv:2604.26529 (2026). (Cited in Supplement S1, §8.4.)

[49] M. Tristram et al., "Cosmological parameters derived from the final Planck data release (PR4)," A&A 682, A37 (2024). (Cited in Supplement S3; not used in the dataset.)

[50] E. Tiesinga, P.J. Mohr, D.B. Newell, B.N. Taylor, "CODATA recommended values of the fundamental physical constants: 2022," Rev. Mod. Phys. 97, 025002 (2025). (Cited in Supplement S3.)

[51] V. V. Nikulin, "Integer symmetric bilinear forms and some of their geometric applications," Izv. Akad. Nauk SSSR Ser. Mat. 43 (1979); English transl. Math. USSR Izv. 14 (1980).

[52] Belle II Collaboration, "Measurement of the τ-lepton mass with the Belle II experiment," Phys. Rev. D 108, 032006 (2023), doi:10.1103/PhysRevD.108.032006, arXiv:2305.19116.

[53] S. K. Donaldson, "Adiabatic limits of co-associative Kovalev–Lefschetz fibrations," in Algebra, Geometry, and Physics in the 21st Century, Progr. Math. 324, Birkhäuser (2017), pp. 1–29, arXiv:1603.08391.

[54] C. Camere, "Lattice polarized irreducible holomorphic symplectic manifolds," Ann. Inst. Fourier 66 (2016), 687–709, arXiv:1312.3224.

[55] C. Zhou, Z. Zhou, "Algebraic Stability and Cosmological Structure E: The Arithmetic Necessity of Three Generations of Fermions," Zenodo, doi:10.5281/zenodo.18626667 (2026).

[56] Particle Data Group, S. Navas et al., "Review of Particle Physics," Phys. Rev. D 110, 030001 (2024), doi:10.1103/PhysRevD.110.030001 (cited in S3 only)

[57] CDF Collaboration, T. Aaltonen et al., "High-precision measurement of the W boson mass with the CDF II detector," Science 376, 170 (2022), doi:10.1126/science.abk1781

[58] JUNO Collaboration, "First measurement of reactor neutrino oscillations at JUNO," arXiv:2511.14593 (2025).

[59] JUNO Collaboration, "Sub-percent precision measurement of neutrino oscillation parameters with JUNO," Chin. Phys. C 46, 123001 (2022), arXiv:2204.13249.

[60] D. Buttazzo, G. Degrassi, P.P. Giardino, G.F. Giudice, F. Sala, A. Salvio, A. Strumia, "Investigating the near-criticality of the Higgs boson," JHEP 12 (2013) 089, arXiv:1307.3536.

[61] DESI Collaboration, M. Abdul Karim et al., "DESI DR2 results. II. Measurements of baryon acoustic oscillations and cosmological constraints," Phys. Rev. D 112, 083515 (2025), arXiv:2503.14738.